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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

EC3224-1 – EMBEDDED C PROGRAMMING USING PSoC
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. What is the benefit of using ``typedef`` in embedded systems?
A) Reduces execution time
B) Allocates memory dynamically
C) Improves code readability and portability
D) Compiles faster
2. Which qualifier prevents compiler optimization of a variable?
A) `static`
B) `volatile`
C) `const`
D) `extern`
3. A variable defined with ``const`` means:
A) Value cannot be modified after initialization
B) Variable is always global
C) Variable must be stored in ROM
D) Compiler ignores it
4. The keyword ``extern`` in C is used for:
A) Defining a global variable
B) Declaring a variable defined in another file
C) Making variable constant
D) Allocating dynamic memory
5. Which of the following about ``static`` keyword is true?
A) Inside function → preserves value between calls
B) Outside function → restricts variable to file scope
C) Both A and B
D) None
6. Implicit type casting in C is also called:
A) Casting
B) Overloading
C) Type demotion
D) Type promotion
7. In C, arguments passed to functions are by default:
A) Pass by reference
B) Global
C) Pass by pointer
D) Pass by value
8. Which of the following correctly declares a function returning an ``int``?
A) `int func();`
B) `func() int;`
C) `function int func;`
D) `int return func();`
9. How do you access members of a structure using a pointer?
A) dot (.) operator
B) arrow (→) operator
C) :: operator
D) Both . and →
10. Which statement about nested structures is correct?
A) A structure cannot contain another structure
B) Nested structures are illegal in C
C) Structures can contain other structures as members
D) Only unions can contain structures

11. Which control structure is best suited for implementing state machines in embedded
 A) switch-case B) if-else
 C) for loop D) do-while
12. If a structure contains another structure, how do you access nested members?
 A) dot chaining (e.g., outer.inner.member) B) with * operator
 C) Cannot be accessed directly D) Only with typedef
13. What is the size of a structure?
 A) Sum of all member sizes B) Depends on alignment and padding
 C) Always multiple of 4 D) Always equal to the largest member
14. Which is the correct way to initialize an array?
 A) `int arr[3] = {1,2,3};` B) `int arr[] = {1,2,3};`
 C) `int arr[] = {1,2,3};` D) None
15. An array of structures means:
 A) Each element is a structure B) Each element is a pointer
 C) Structures cannot be in arrays D) Structures cannot be in arrays
16. Which is the correct syntax for a function pointer to a function returning int and taking no
 A) `int *fptr();` B) `int (*fptr)();`
 C) `(fptr) int();` D) `int fptr();`
17. Which of the following is true about interrupts?
 A) They always stop the CPU permanently B) They cannot be disabled
 C) They replace the main program completely D) They temporarily suspend main program to
18. Nested interrupts mean:
 A) One ISR calls another ISR B) Interrupts cannot occur during another
 C) Multiple ISRs can execute at the same time D) Higher-priority interrupts can preempt lower-priority ISRs
19. UART communication is:
 A) Parallel and synchronous B) Serial and asynchronous
 C) Serial and synchronous D) Only parallel
20. Which protocol is synchronous and supports master-slave with chip select?
 A) UART B) I2C
 C) SPI D) CAN

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) What are the different classifications of embedded system
 b) Write an Embedded C program to perform the following operations on the STATUS_REG register using bitwise operators:
 Turn ON the system power (SYS_POWER) and set the system to Manual Mode (MODE_SEL = 1).
 If the TEMP_SENSOR bit is 1, turn ON the COOLER and STATUS_LED bits.
 If the OVERLOAD bit is 1, turn ON the ALARM and turn OFF the MOTOR immediately

Marks 8 BT* L*2 CO* 1 PO* 1

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
OVERLOAD	ALARM	MOTOR	COOLER	TEMP_SENSOR	MODE_SEL	STATUS_LED	SYS_POWER

Figure : STATUS_REG

8 L2 2 1

2.	a)	Explain the use of if-else, switch-case, and loop constructs (for, while, do-while) in Embedded C.	8	L2	1	1
	b)	Write an embedded C program to find the largest values in a given array. Use a function call <i>Find_Largest(uint8_t *array_address, int *array_size, *largest_value)</i>	8	L3	2	1
3.	a)	Explain pointer declaration, dereferencing, and passing pointers to functions with examples.	8	L2	1	1
	b)	Write an embedded C program to make HEX(ranging from 0x00 to 0xFFFFF) to BCD conversion.	8	L3	2	1
Unit – II						
4.	a)	Discuss how one-dimensional and two-dimensional arrays are stored in memory in embedded systems.	8	L2	3	1
	b)	Design a structure struct Node that includes: - id - struct Sensor { char type[10]; float value; } sensor; Write a function void updateNode(struct Node *n, float newValue) to modify the sensor value via pointer access.	8	L3	4	1
5.	a)	Define a pointer to function and explain its syntax and usage.	8	L2	3	1
	b)	Create an array of 5 structures, each representing a Device with members: device_id, power_status, and voltage. Initialize the array and write a function to calculate the average voltage of all powered devices.	8	L3	4	1
6.	a)	Explain different ways of accessing structure members (using structure variable, pointer, and arrow operator) with example code.	8	L3	3	1
	b)	Simulate a 4x4 LED matrix using a 2D array unsigned char led[4][4]. Write functions to: - Turn on an LED at a given row and column. - Display the entire LED matrix state.	8	L3	4	1
Unit – III						
7.	a)	Describe the functionality of IDE in developing embedded system.	8	L2	5	1
	b)	Provide the functional interconnection of 16x2 LCD interface with PSOC. Provide the required basic API structures required for implementation.	8	L2	5	1
8.	a)	How does a matrix keypad interface with PSOC? Provide the functional flow of programming to detect a key hit.	8	L2	5	1
	b)	What are use of flash memory system in embedded systems? How EEPROM memory is accessed in PSOC?	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
